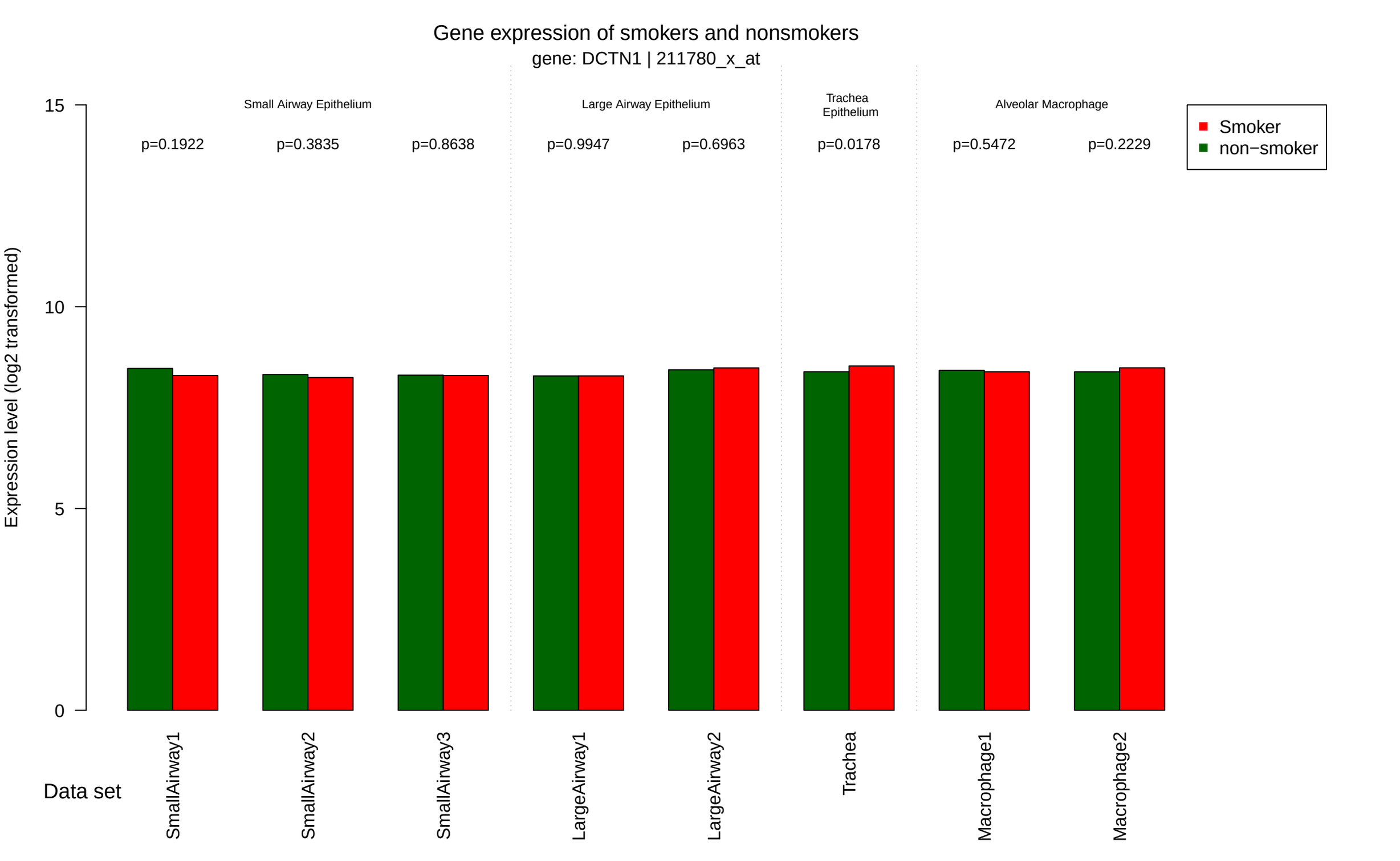
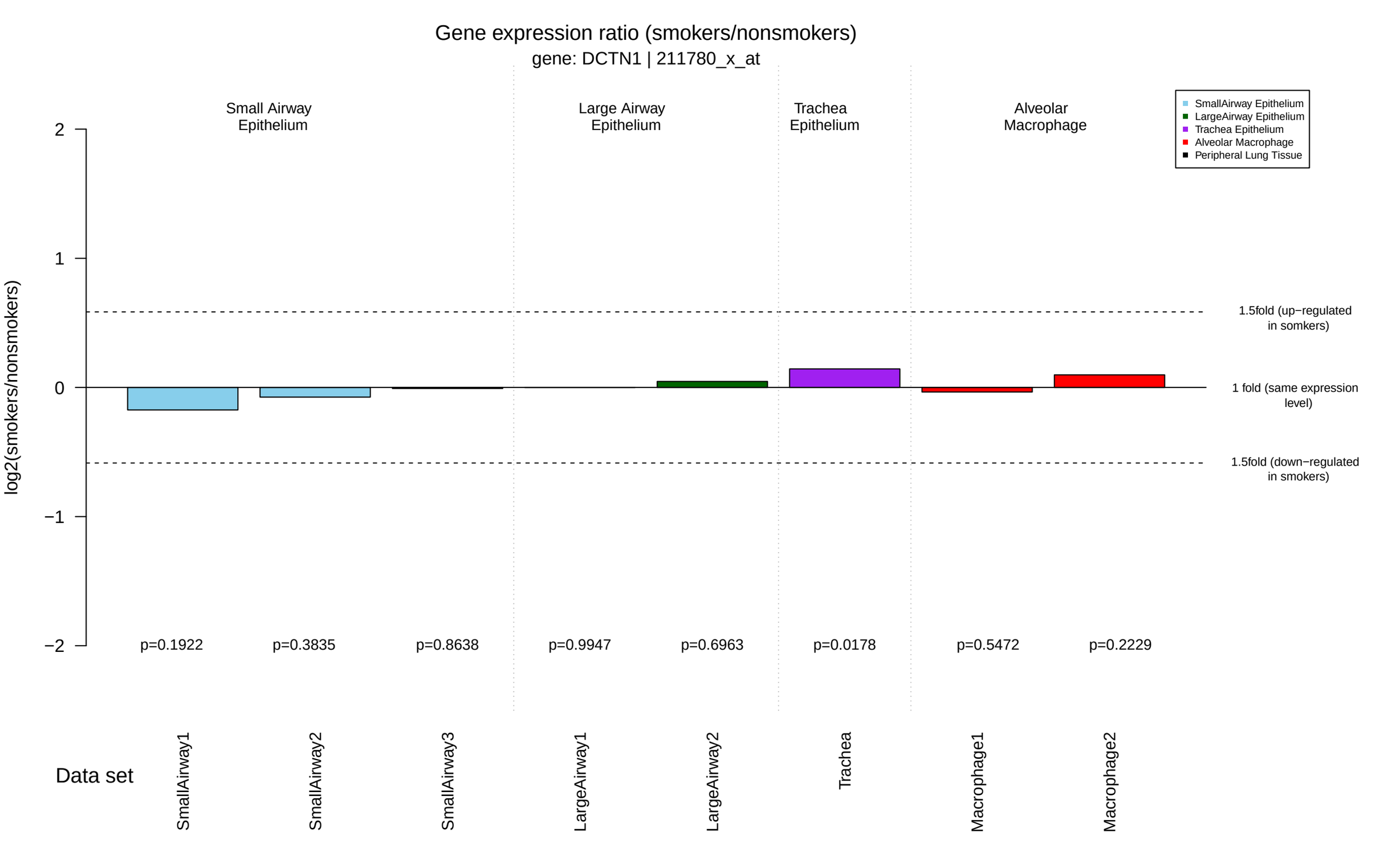
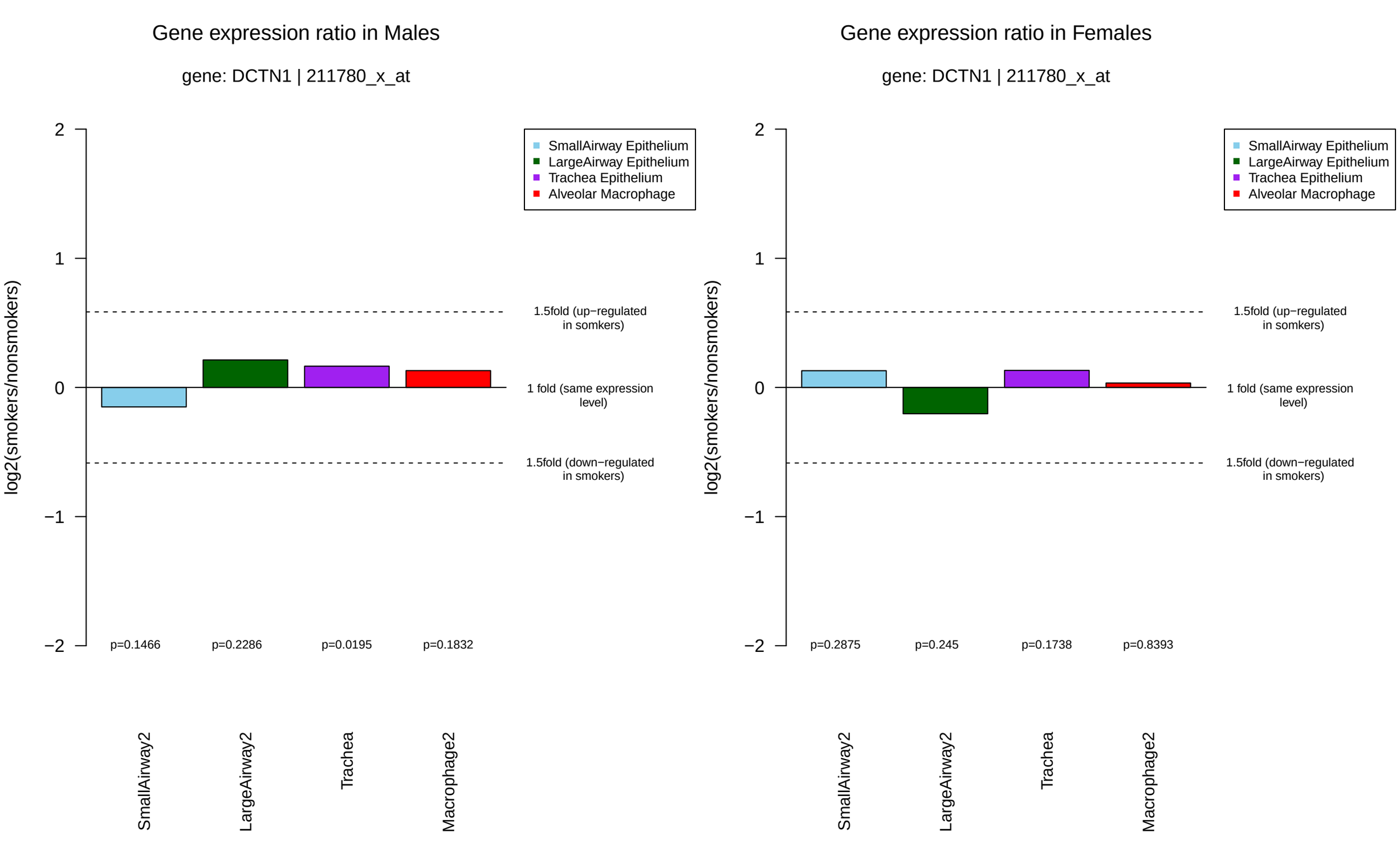
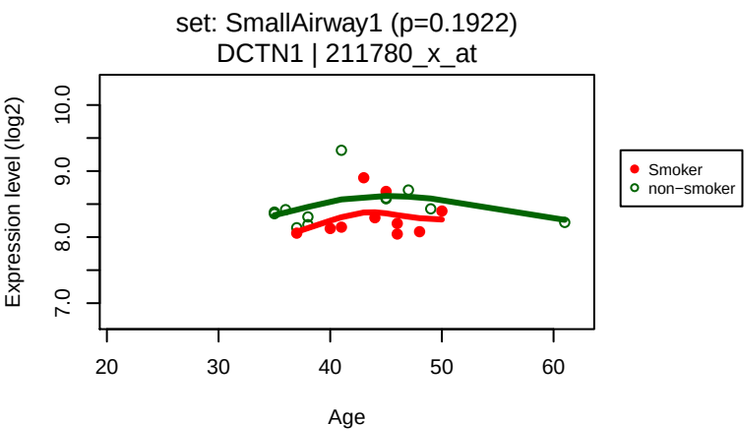
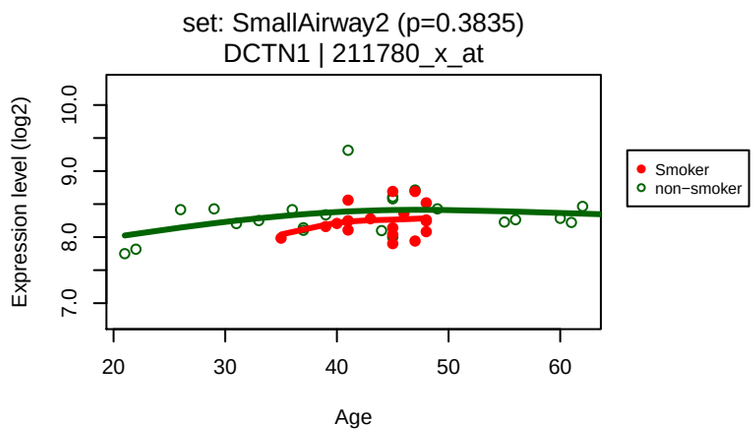
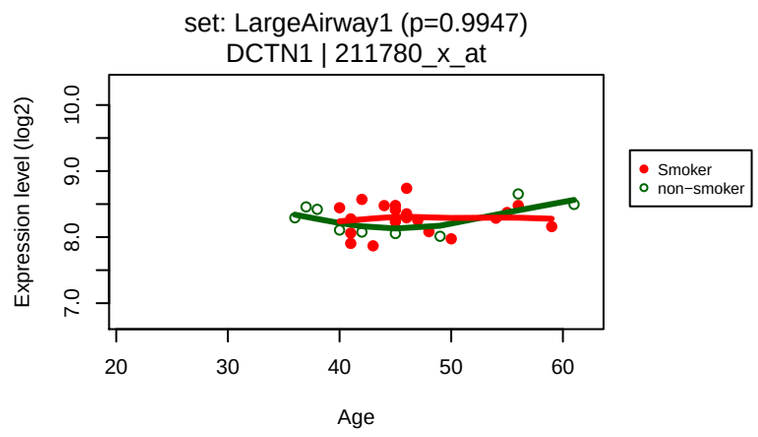

Gene expression of smokers and nonsmokers
gene: DCTN1 | 211780_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1922
p=0.3835
p=0.8638
p=0.9947
p=0.6963
p=0.0178
p=0.5472
p=0.2229
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

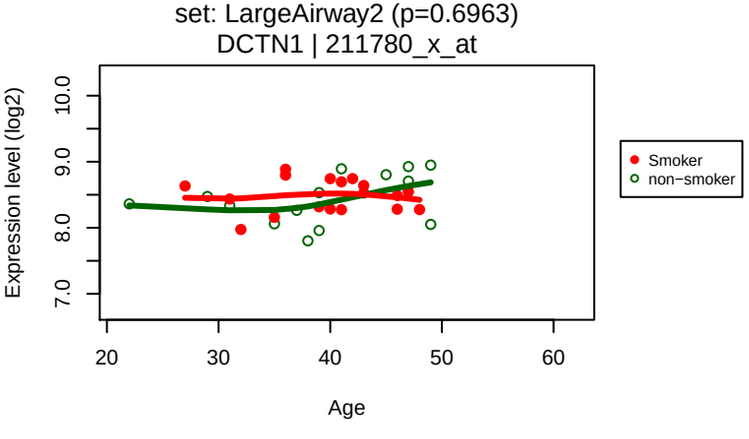

Gene expression ratio (smokers/nonsmokers)
gene: DCTN1 | 211780_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1922
p=0.3835
p=0.8638
p=0.9947
p=0.6963
p=0.0178
p=0.5472
p=0.2229
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

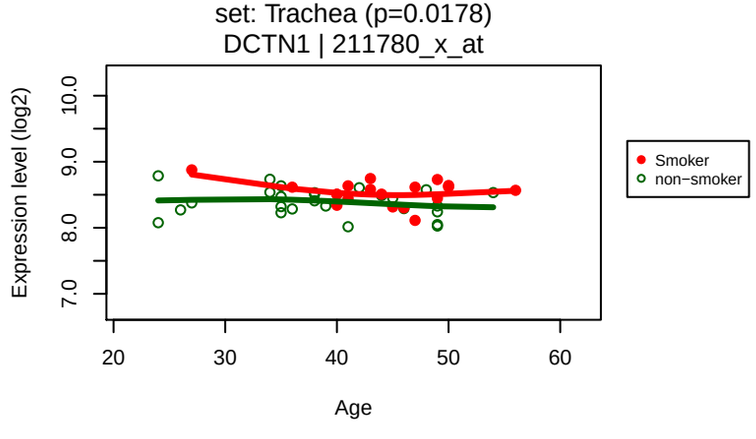

Gene expression ratio in Males
gene: DCTN1 | 211780_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1466
p=0.2286
p=0.0195
p=0.1832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DCTN1 | 211780_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2875
p=0.245
p=0.1738
p=0.8393
SmallAirway2
LargeAirway2
Trachea
Macrophage2

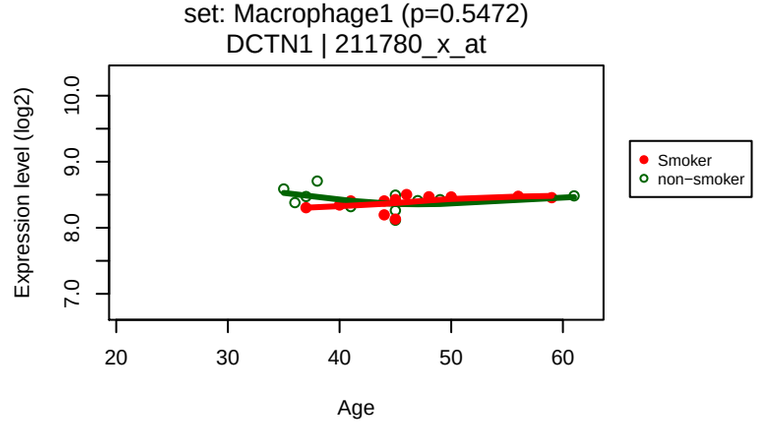

set: SmallAirway1 (p=0.1922)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

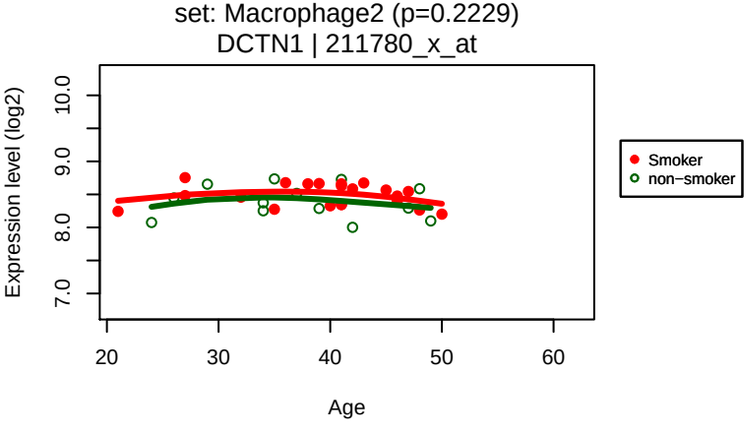

set: SmallAirway2 (p=0.3835)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

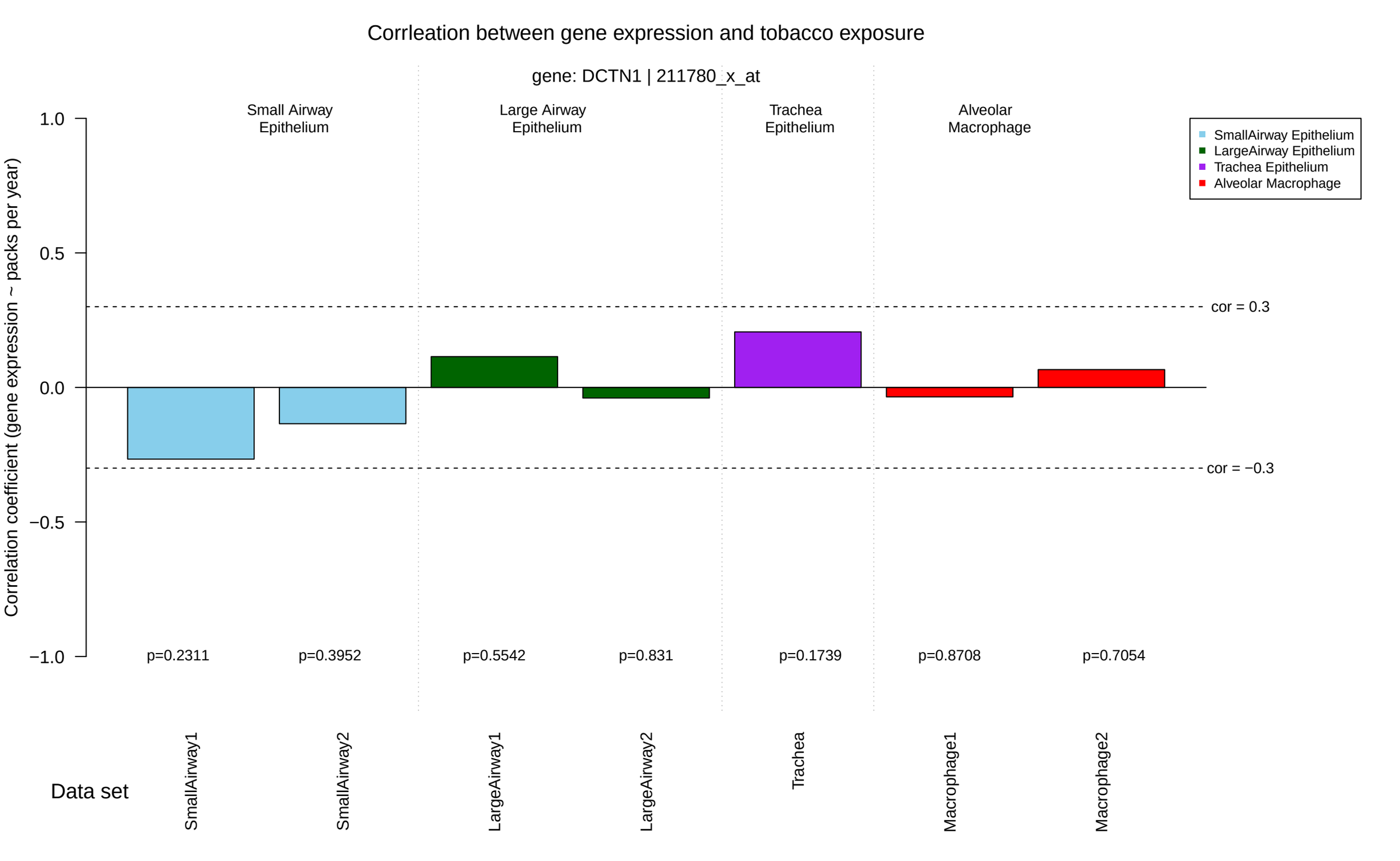

set: LargeAirway1 (p=0.9947)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6963)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0178)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5472)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2229)
DCTN1 | 211780_x_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DCTN1 | 211780_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2311
p=0.3952
p=0.5542
p=0.831
p=0.1739
p=0.8708
p=0.7054
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2